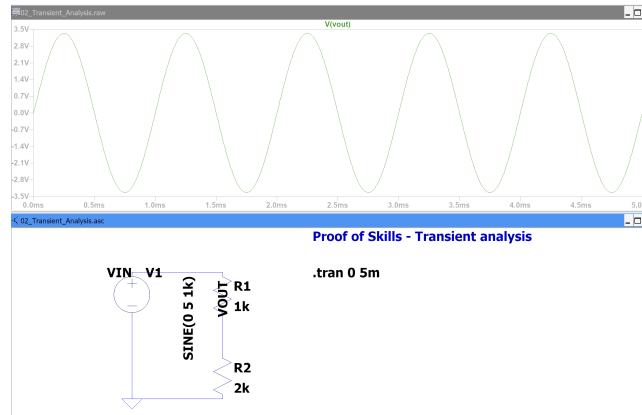


LTspice Data Export and MATLAB Import

LTspice transient data was exported as text and read in MATLAB. The source waveform, MATLAB import script, and resulting comparison plot are included below.



LTspice transient waveform used as the exported simulation data.

```
MATLAB Drive/Activity_04_LtSpice.m
3
4 clear;
5 close all;
6 clc;
7
8 script_dir = fileparts(mfilename('fullpath'));
9 if isempty(script_dir)
10     script_dir = pwd;
11 end
12
13 data_file = fullfile(script_dir, 'data', 'transient_export.txt');
14 if ~isfile(data_file)
15     data_file = fullfile(pwd, 'transient_export.txt');
16 end
17 if ~isfile(data_file)
18     error('Cannot find transient_export.txt. Upload the data file or ', ...
19         'the complete MATLAB folder, then run this script again.');
```

```
21
22 table_data = readtable(data_file, 'FileType', 'text', 'Delimiter', '\t');
23 time_s = table_data.time;
24 V_VIN = table_data.V_VIN;
25 V_VOUT = table_data.V_VOUT;
26
27 gain = (V_VIN + V_VOUT) / (V_VIN * V_VIN);
28 expected_gain = 2/3;
29
30 figure('Color', 'w');
31 plot(time_s*1e3, V_VIN, 'Color', [0.15 0.45 0.85], 'LineWidth', 1.8);
32 hold on;
33 plot(time_s*1e3, V_VOUT, 'Color', [0.85 0.25 0.15], 'LineWidth', 1.8);
34 grid on;
35 xlabel('Time (ms)');
36 ylabel('Voltage (V)');
37 title('LTspice Transient Export Imported into MATLAB');
```

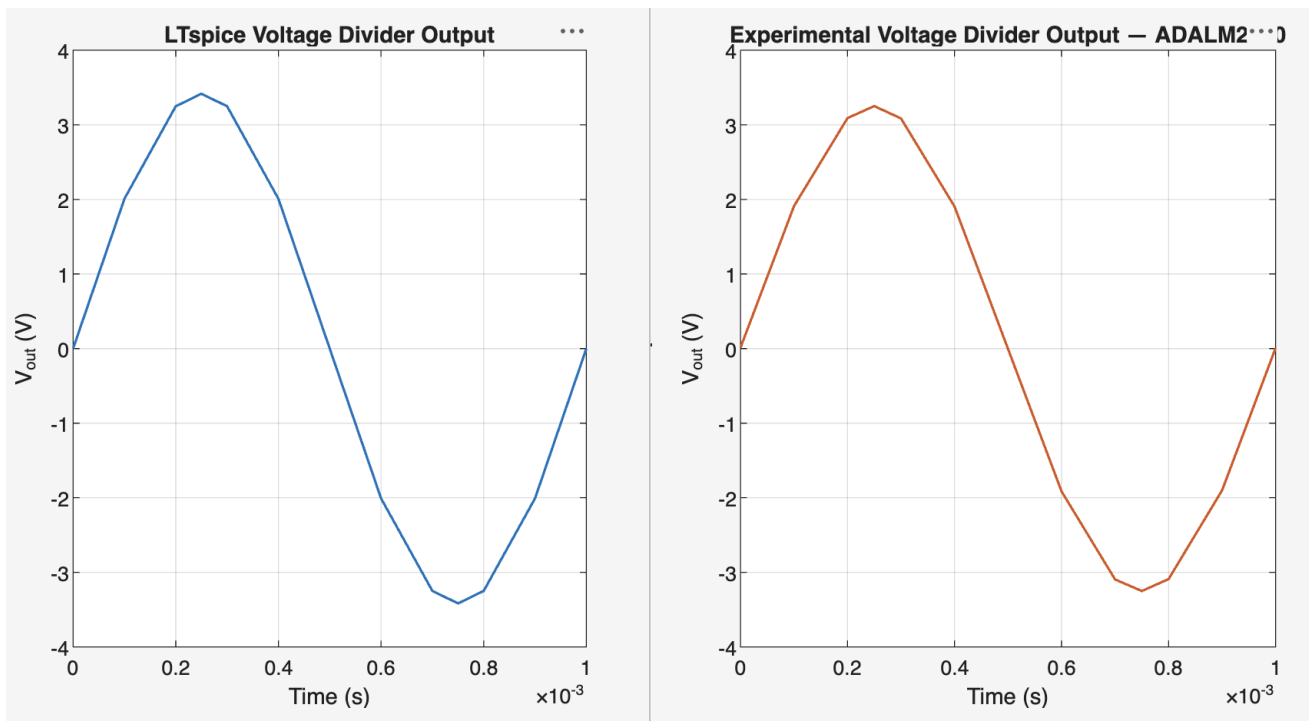
Command Window

```
Imported 201 samples from /MATLAB Drive/transient_export.txt
Measured divider gain VOUT/VIN: 0.666666667
Expected divider gain: 0.666666667
>> |
```

MATLAB import code and Command Window confirmation.

MATLAB Waveform Comparison

MATLAB comparison plot of the LTspice voltage-divider output and experimental ADALM2000 output.



MATLAB figure comparing the simulated and experimental waveforms.